



Munich University of Applied Sciences

Hochschule Muenchen

Department of Computer Science & Mathematics

Fakultaet fuer Informatik und Mathematik

Aria-NBV: Quality-Driven Next-Best-View Planning with Egocentric Foundation Models

TODO: Deutscher Titel

Master's Thesis Proposal / Proposal zur Masterarbeit

for obtaining the academic degree Master of Science (M.Sc.)

submitted to
Munich University of Applied Sciences
Department of Computer Science & Mathematics

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Submission Date: TODO: Abgabedatum

Transparency in the Use of AI Tools

TODO: Document AI tools used while preparing the proposal according to the final institutional requirements.

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1 Motivation

NBV planning is central to active 3D reconstruction because each additional view consumes acquisition budget. ARIA-NBV proposes to rank candidate egocentric views by expected reconstruction-quality gain rather than by coverage alone.

2 Problem Statement

The core problem is to predict and plan with RRI signals for egocentric indoor scenes in ASE. The proposal should make the target setting, candidate action space, reconstruction-quality metric, and evaluation budget explicit before expanding to non-myopic planning.

3 Objectives

TODO: State the thesis research questions and measurable objectives. Keep the current seminar-paper scope as the baseline and separate required thesis contributions from optional extensions.

4 Proposed Method

TODO: Summarize the oracle RRI labeler, VIN scorer, calibration checks, and planned comparison between greedy and longer-horizon view selection.

5 Schedule

TODO: Add the thesis timeline, milestones, risks, and fallback scope.

6 Preliminary Thesis Outline

TODO: Add the planned thesis chapter structure. The HM guide expects the proposal to include a rough outline, an initial project plan, and a preliminary bibliography; the bibliography is rendered at the end of this proposal.

Bibliography